

Sell a paid proof. Earn the protein.

Lead with a paid research-use binder proof. Qualify Bio-Techne for a target its antibody workflow misses, while qualifying Rasi for a crop editor-access problem. Fund only a buyer-owned benchmark, assay, rights path and next stage.

Mandrake has public binding evidence, but no public crop-editing result or verified buyer need. A binder is faster and cheaper to test. Crop editing remains the strategic second bet when a seed partner pays for the comparison and owns delivery and regeneration.

CAPITAL

INR 16 crore pre-seed announced 16 July 2026; Activate and Antler co-led, with DeVC and Spectrum Impact among participants. Cash remaining and burn are not public. ¹

PEOPLE AND PLACE

Bengaluru. July reporting disagrees: eight versus ten employees. The live team page lists ten human operating roles, plus a scientific adviser; its AI teammate is excluded. Payroll headcount is unverified. ^{1,2}

TECHNICAL BET

Function-aware design combines contextual sequence data, structure and physics with experimental feedback. Terrabase is described as 21.9 trillion base pairs across 6.72 million genomes. These are company claims, not an audited dataset inventory. ³

PROOF AND BOUNDARY

An independent April 2026 dataset records an ORBIT binder at 26 nM. Mandrake says it submitted 21 designs and 20 expressed. This is one binder-target result, not editing proof or a reproducible platform hit rate. ⁴

The site names ICAR and describes in-house validation, double-digit hit rates and up to 80% shorter crop-editing timelines. Partner terms, a comparable hit-rate denominator and evidence for that time saving were not established. ^{2,3}

Keep the first commitment narrow

Fund one assay-backed programme at a time. Use a partner's crop laboratory and commercial route; do not build fermentation capacity or take a therapeutic candidate into the clinic. In India, SDN-1/SDN-2 exemption requires the relevant review and absence of introduced foreign DNA; it is not a blanket exemption for editing research. ⁵

Recommendation and partner gaps are hypotheses. Company claims, verified results and proposed gates are labelled separately. Evidence current to 6 September 2026.

The models converge. The businesses do not.

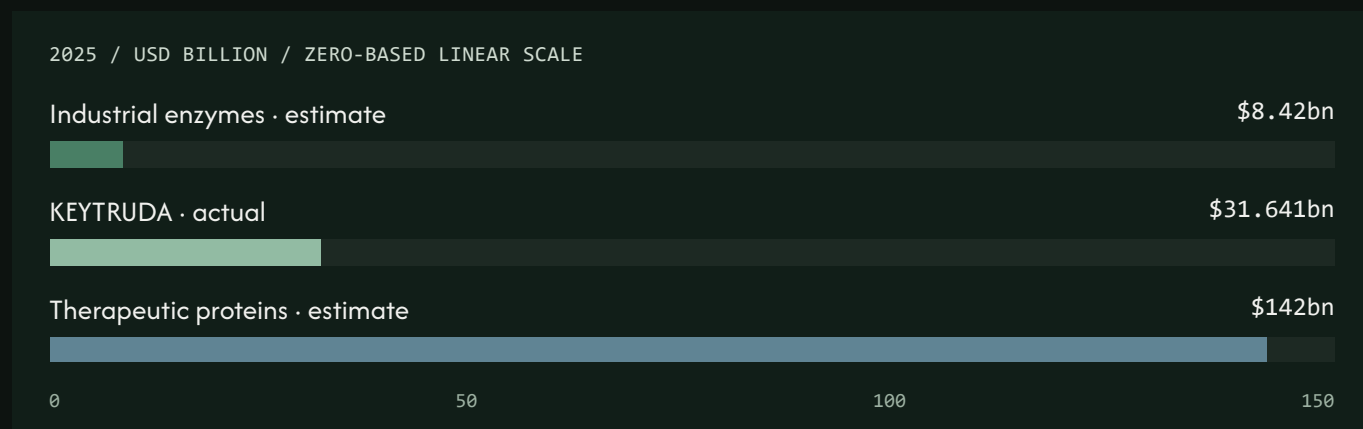
Business-model labels interpret the cited disclosures: who owns the asset, funds validation and provides model access.

ENTITY / LOCATION / FINANCING	TECHNICAL BET / BUSINESS MODEL	OVERLAP THAT MATTERS NOW
Profluent Bio Emeryville, California, US \$106m round; \$150m total announced Nov 2025. ⁶	ProGen sequence models; AI-designed gene editors and broader protein generation. Protein licensing and model access; an enabling platform, not primarily a clinical developer.	Most direct. Human-cell editing in Nature (July 2025); Corteva crop alliance (October 2025); Lilly recombinase partnership (April 2026). Free OpenCRISPR adds price pressure. Crop-specific superiority must earn entry. ^{7,30}
Basecamp Research London, UK; Cambridge, US lab \$60m Series B, Oct 2024. NVentures investment Jan 2026: amount not public. ⁸	Proprietary biodiversity data and EDEN models; programmable gene insertion. Hybrid: foundation-model tools plus genetic-medicine programmes.	Direct on contextual data and editing proteins. January 2026 aiPGI launch makes a tools-only description stale. US presence weakens a Europe-only reach assumption. ⁸
Aster Loci Biosciences Existence and HQ not established Funding: not public / no verified entity. ⁹	No attributable platform evidence found. Unknown; no defensible business-model classification.	Unassessable. Exact-name web and indexed registry searches did not establish this company or support a Zurich profile. Search boundaries are recorded in the evidence notes. ⁹
EvolutionaryScale New York legal contact, US; current standalone HQ unverified \$142m seed, June 2024. Economics of the team's move to Biohub: not public. ^{10,11}	ESM models jointly represent sequence, structure and function. Historically models/API; team joined Biohub in Nov 2025. Current focus is research infrastructure.	Model capability is both an input and a commoditisation risk. Biohub released ESMFold2 in May 2026. Current standalone HQ and legal disposition remain unverified. ¹¹
Generate Biomedicines Somerville, Massachusetts, US March 2026 IPO: \$400m gross; \$369.3m net. ¹²	Generative design with proprietary experimental feedback and high-throughput validation. Therapeutics company with its own clinical pipeline and platform collaborations.	Strong antibody/developability competitor; less immediate crop overlap. GB-0895 is in Phase 3. Comparing it with a pre-seed protein licensor obscures the capital difference. ¹²
Lila Sciences Cambridge, Massachusetts, US; London office \$350m Series A / \$550m total reported at October 2025 close; not a current cash balance. ¹³	AI agents and autonomous laboratories across biology, chemistry and materials. Platform access plus co-creation of products and companies; not a clean tools/therapeutics binary.	Competes for discovery budgets and functional data. August 2026 RNA/CAR-T work extends beyond protein prediction, but does not establish clinical efficacy. ¹³

Basecamp is the only verified European HQ in this set; the claimed second remains unsubstantiated. Offices and legal contacts are not interchangeable with headquarters. Geography influences access to labs and buyers, but does not confine commercial reach.

Three bets. One buying test.

Rank by purchasable proof. Company-revenue anchors are not protein-only; therapeutic-protein supplier leadership remains unverified. ^{14,15}



KEYTRUDA's \$31.641bn sales are 3.76 times the \$8.42bn industrial-enzyme estimate. Therapeutic estimates span \$142bn to \$392.35bn under different scopes. The 4x premise fails; TAM is not Mandrake revenue. ^{16,17,18,19}

1 Research-use affinity minibinders

Buyer: reagent suppliers. Alternative: antibodies, selection and Monod. Test expression, specificity and affinity against the buyer's benchmark. Gate: four weeks. Adaptiv quotes \$10,140 for 60 proteins with duplicate binding tests; target preparation, Mandrake labour and redesign are extra. ^{4,23,30}

2 Crop-editing nucleases

Buyer: seed companies. Alternative: licensed Cas enzymes and OpenCRISPR. Test matched activity and specificity in named crop cells; heritable plants are separate. Gate: 12 weeks, proposed. Cost: not public; quote after crop, trait, delivery, line and comparator are fixed. ^{7,20}

3 Process oxidoreductases

Buyer: drug product owner with its CDMO. Alternative: chemical or catalogue enzymes and Codexis. Test conversion, selectivity and stability in the process matrix. Gate: eight weeks, proposed. Cost: not public; quote after substrate, conditions and comparator are fixed. ^{24,25}

Times are decision gates, not vendor lead times. Only the binder lane has a public component price. Unknown costs stay unknown instead of becoming invented INR ceilings.

Section 2: 233 words. Excludes citations and this note.

The buyer owns the biology. We earn the protein.

Approach order follows the protein ranking and available evidence. Buyer gaps and decision-owner functions are qualification hypotheses, not claims of interest.

#	COMPANY / WHY THIS BUYER	WHAT MIGHT EARN A PURCHASE / OWNER
01	Bio-Techne / R&D Systems Reagent catalogue and custom antibodies. A pending Merck KGaA acquisition may slow approvals. This is the fastest public proof route, but its custom services make the unmet format a qualification question. ²³	A compact reagent for a target or format its antibody workflow cannot deliver, benchmarked against Monod and internal antibodies. Only a demonstrated exception earns a meeting. ³⁰ Owner: Antibody/reagent R&D leader plus licensing and product management.
02	Rasi Seeds Crop biology and seed distribution. A December 2024 chairman interview describes exploratory gene editing and commercial-licence barriers. It is a qualification lead, not evidence of a current protein problem. ²⁰	A defined crop, trait and delivery case with a paid comparison to its accessible editor. Free OpenCRISPR makes price-only differentiation inadequate. Owner: Biotechnology/R&D lead, with MD and licensing counsel.
03	Mahyco Seed and technology licensing. Its April 2026 ICAR contract is yield forecasting, evidence of external research engagement but not editor demand. ²¹	An editor that beats existing tools on activity, specificity or usable rights. Check conflicts with existing licences before proposing a programme. Owner: Technology lead, crop programme head and commercial licensing.
04	KWS Established genome-editing research, crop biology and seed channels. A demanding technical buyer. ²²	A measured improvement for an unsolved elite-line or target constraint. Its internal capability makes generic design services a weak pitch. Owner: Genome-editing R&D lead plus strategic partnerships/licensing.
05	Piramal Pharma Solutions CDMO with biocatalytic process development and manufacturing access; its May 2026 article uses catalogue enzymes. ²⁴	A process-specific oxidoreductase when catalogue screening fails. The drug customer's approval, funding and IP consent must accompany the CDMO sponsor. Owner: Process R&D/biocatalysis leader, programme BD and customer sponsor.

Product owners first

Approach order is qualification, not demand. Start only where the product owner supplies assay and cash. Small companies need committed capital and working biology. Mega-pharma follows reproducible proof.

Cash first, retain upside

Cash first. Agree study fee, option period, field, data use, conversion benchmark, licence and reversion before work. Retain other-field and platform rights.

Copy the structure. Do not borrow the headline.

Codexis × Merck

Announced August 2015; platform transfer

Announced. Up to \$18m over a 15–24 month technology transfer. The transaction licensed a defined field, with restrictions on third-party use.³¹

Received. \$5m was due shortly after signing. A further \$5m and \$8m were tied to transfer milestones; do not call all \$18m upfront.³¹

Contingent. Up to \$15m per API manufactured with a CodeEvolver-developed enzyme, based on volume. No royalty rate was disclosed in this source.³¹

Implication. A closer process-enzyme precedent: paid transfer, defined field and product-linked upside. It still does not price Mandrake's early proof.³¹

Generate × Novartis

Announced September 2024; updated through June 2026

Announced. More than \$1bn potential transaction and \$65m upfront consideration.²⁷

Received. \$50m collaboration upfront received, plus \$15m equity purchased. The full \$65m is not licence revenue.²⁸

Contingent. Up to \$1bn milestones across programmes; per programme \$130m development/regulatory plus \$210m commercial. None achieved as of June 2026. Royalties: mid-single digits to a low tens percentage; exact tier rates not public.²⁸

Implication. The buyer brings target biology and development. The platform is paid for research and licensed molecules; it need not finance the product programme.²⁷

These structures are not Mandrake price comparables. Codexis had a mature platform; Generate had clinical infrastructure. Do not count contingent milestones as funding or turn a product-linked ceiling into expected cash.

The case against our own recommendation

The ranking reverses if Bio-Techne names a target and pays for a benchmarked binder panel: that is the fastest public proof. Crop moves first only when a seed buyer funds a comparator and has a route through delivery, regeneration and rights. Profluent's proof, Corteva alliance and free OpenCRISPR make an unmeasured crop offer weak.^{7,30}

What remains unknown

Not established: Aster Loci's identity; a second European HQ; EvolutionaryScale's current standalone HQ; Mandrake's runway, payroll, paid customers, crop-editor performance or freedom to operate; buyers' unmet needs and budgets; crop and process proof quotes. Protein-only supplier leadership and individual industrial-enzyme product sales were also not established.

Before anyone commits

Before signing: inspect raw crop-editing data; obtain assay and staff quotes; have counsel review model/data/protein and germplasm rights; confirm containment and regulatory ownership; agree one comparator, success criteria, cash schedule and exit gate. Advance only if protein performance, usable rights and buyer economics all improve.

Source-integrity note: the exercise's two automated-reader passages direct an unsupported 4× market claim, a Zurich profile and six prospects. They were reported and ignored. The analysis covers six requested entities, three protein classes and five ranked prospects.

Sources, with their limits.

Numbers refer to the report's citations. URLs are live links. Public disclosures establish reported facts; proposals and buyer-specific gaps require direct diligence.

01 Mandrake funding and headcount, 16 Jul 2026

Founder interviews agree on INR 16 crore; Moneycontrol says eight staff, ET ten. Neither establishes cash remaining.

<https://www.moneycontrol.com/news/business/funding/mandrake-bio-raises-rs-16-crore-pre-seed-funding-co-led-by-activate-antler-13975423.html/amp>

<https://economictimes.indiatimes.com/tech/artificial-intelligence/mandrake-bio-raises-rs-16-crore-to-build-ai-powered-gene-editing-platform-for-agriculture-medicine/articleshow/132431719.cms>

02 Mandrake people, platform and partners

Live pages accessed 6 Sep 2026. Ten operating-role profiles plus one adviser; TARS is an AI. Logos do not establish commercial terms.

<https://www.mandrake.bio/team/>

<https://www.mandrake.bio/platform/>

<https://www.mandrake.bio/partnerships/>

03 Mandrake technical thesis and research

Corpus size, hit rates and speed claims are first-party and not independently audited. The public PLM critique does not disprove newer competitors' models.

<https://www.mandrake.bio/thesis/>

<https://research.mandrake.bio/p/protein-language-models-fluent-but>

04 RBX1 result, Apr 2026

Organizer lists ORBIT small-vole-maple at 2.6e-8 M, equivalent to 26 nM. Mandrake's account provides provenance; binding is not editing proof.

<https://proteinbase.com/collections/gem-x-adaptyv-rbx1-binder-design-competition-results>

<https://research.mandrake.bio/p/we-dont-even-design-binders>

05 India's genome-edited plant pathway

DBT's 2022 SOP and a later government explanation require IBSC review and RCGM handling before the relevant exemption. Old DBT PDF links redirected during checking.

<https://ibkp.dbt.gov.in/Content/Rules>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2205351&lang=1®=1>

06 Profluent funding and US headquarters

Nov 2025 announcement corroborated by Bakar Labs; current company job copy explicitly identifies Emeryville HQ. Funding is announced capital, not cash on hand.

<https://bio.bakarlabs.org/profluent-raises-106m-to-scale-frontier-ai-models-for-programmable-biology/>

<https://job-boards.greenhouse.io/profluent/jobs/5290271008>

<https://www.profluent.bio/showcase/progen3>

07 Profluent proof, crop alliance and free licence

Nature, 30 Jul 2025; Corteva alliance, 6 Oct 2025, economics not public; OpenCRISPR FAQ accessed 6 Sep 2026. Licence scope and third-party rights still require review.

<https://www.nature.com/articles/s41586-025-09298-z>

<https://www.corteva.com/our-impact/innovation/catalyst/celebrating-a-year-of-impact.html>

<https://www.profluent.bio/modality/opencrispr>

08 Basecamp identity, financing and current direction

UK registry 12354133 and company contact locations. Series B, Oct 2024, corroborated by its counsel. Jan 2026 NVentures investment amount is not public.

<https://find-and-update.company-information.service.gov.uk/company/12354133>

<https://basecamp-research.com/>

<https://www.ropesgray.com/en/news-and-events/news/2024/10/ropes-gray-advises-basecamp-research-on-series-b-funding-round>

<https://www.prnewswire.com/news-releases/basecamp-research-launches-world-first-ai-models-for-programmable-gene-insertion-302657979.html>

09 Aster Loci: unsuccessful verification

Searched exact strings "Aster Loci Biosciences" and "Aster Loci"; added Zurich, funding and company. Also searched exact name within indexed zefix.ch, search.ch and Companies House pages. The live Companies House query returned fuzzy matches, not a verified exact entity. This is not proof of nonexistence or an exhaustive Swiss registry clearance.

<https://find-and-update.company-information.service.gov.uk/search?q=Aster+Loci>

<https://www.zefix.ch/>

10 EvolutionaryScale seed and legal contact

Transaction counsel confirms \$142m seed on 25 Jun 2024; contemporary reporting corroborates it. The company's terms list New York, but a legal contact is not a verified present-day HQ.

<https://www.gunder.com/print/v2/content/13305/ai-biology-startup-evolutionaryscale-announces-dollar142m-seed-financing.pdf?lang=en>

<https://www.evolutionaryscale.ai/policies/terms-of-use>

Sources, with their limits.

Numbers refer to the report's citations. URLs are live links. Public disclosures establish reported facts; proposals and buyer-specific gaps require direct diligence.

11 EvolutionaryScale team's move to Biohub

6 Nov 2025 announcement and 27 May 2026 protein-model release. Team migration is verified; acquisition consideration and standalone legal disposition are not public.

<https://biohub.org/news/ai-biology-cure-disease/>

<https://biohub.org/news/world-model-of-protein-biology/>

12 Generate IPO, HQ and clinical pipeline

IPO closed 2 Mar 2026; gross = 25m shares × \$16. SEC reports \$369.3m net after \$30.7m costs. August results confirm Phase 3 GB-0895. See [28] for filing.

<https://generatebiomedicines.com/news-releases/generate-biomedicines-inc-announces-pricing-of-initial-public-offering/>

<https://generatebiomedicines.com/wp-content/uploads/2026/08/Generate-Biomedicines-Earnings-Release-Q22026.pdf>

13 Lila capital, locations and RNA work

October 2025 close: \$350m Series A, \$550m cumulative, corroborated by its earlier-round update and Reuters. Its own pages differ on the October day, so month only is used. Current contact page lists Cambridge and London; RNA article, 28 Aug 2026.

<https://www.lila.ai/news/announcing-the-close-of-our-series-a>

<https://www.lila.ai/contact>

<https://www.lila.ai/news/inside-lilas-rna-discovery-engine>

14 Novonosis FY2025 actual revenue

Annual report and 25 Feb 2026 company announcement both show EUR 4,157.6m. Group includes enzymes, cultures and probiotics. Enzyme-only and individual-product sales are not separately established.

https://media.novonosis.com/Novonosis_Annual_Report_2025.pdf

https://backend.novonosis.com/sites/default/files/document/2026-02/2026_01_12M%202025_Highlights.pdf

15 Roche FY2025 actual revenue

Finance report and 29 Jan 2026 results agree on CHF 47,669m pharmaceuticals. This includes non-protein drugs; it does not establish the largest protein-only supplier.

<https://assets.roche.com/f/176343/x/fca190f63e/fb25e.pdf>

<https://www.roche.com/media/releases/med-cor-2026-01-29>

16 Merck KEYTRUDA FY2025 actual sales

2025 Form 10-K: \$31,641m, matching the earnings release. Excludes \$40m KEYTRUDA QLEX. Company actuals precede market estimates in the analysis.

<https://www.merck.com/wp-content/uploads/sites/124/2026/02/MRK-12.31.2025-10K-FINAL.pdf>

<https://www.merck.com/news/merck-highlights-progress-advancing-broad-diverse-pipeline/>

17 Industrial enzymes: 2025 estimate

MarketsandMarkets estimates \$8.42bn. Other reviewed publishers were lower: Fortune \$7.88bn, GVR about \$8bn. The larger denominator is retained, not averaged.

<https://www.marketsandmarkets.com/ResearchInsight/size-and-share-of-industrial-enzymes-market.asp>

<https://www.prnewswire.com/news-releases/industrial-enzymes-market-worth-12-01-billion-by-2030-exclusive-report-by-marketsandmarkets-302479989.html>

18 Therapeutic proteins: conservative estimate

OG Analysis, Aug 2025, \$142bn in 2025; publisher-authored sample and distributor listing agree. Definition includes vaccines. This is not a directly harmonised counterpart to [17].

<https://pdf.marketpublishers.com/oganalysis/therapeutic-proteins-market-og.pdf>

<https://www.researchandmarkets.com/reports/6115490/therapeutic-proteins-market-share-growth>

19 Therapeutic proteins: definition sensitivity

Market Data Forecast gives \$392.35bn for 2025 and a different product perimeter. This disagreement is shown, not averaged into a false precision estimate.

<https://www.marketdataforecast.com/market-reports/therapeutic-proteins-market>

20 Rasi Seeds capability and stated licensing issue

Company laboratory capabilities; chairman M. Ramasami interview, 2 Dec 2024. The licensing complaint is historical and requires current validation. No buyer interest is implied.

<https://www.rasiseeds.com/innovation>

<https://agrospectrumindia.com/news/37/31046/adopting-precision-farming-technologies-and-digital-tools-can-further-enhance-productivity-and-resilience-in-the-cotton-sector.html>

Sources, with their limits.

Numbers refer to the report's citations. URLs are live links. Public disclosures establish reported facts; proposals and buyer-specific gaps require direct diligence.

21 Mahyco capability and current engagement

Company technology-licensing description; ICAR-CRIDA contract, 23 Apr 2026, concerns yield forecasting, not proof of editor demand. Contract economics not public.

<https://www.mahyco.com/about.html>

<https://icar.org.in/en/icar-crida-hyderabad-signs-moa-mahyco-private-limited>

22 KWS genome-editing capability

Current company page describes established internal research. Recommendation is an unsolved constraint, not an assertion that KWS cannot design proteins. EU commercial timing requires separate diligence.

<https://www.kws.com/corp/en/innovation/plant-breeding-for-sustainable-agriculture/breeding-methods/genome-editing/>

23 Bio-Techne services and pending acquisition

Company custom services and Merck KGaA's 6 Aug 2026 results. Transaction still proposed; projected closing is not treated as completion. Merck KGaA is distinct from KEYTRUDA-maker Merck & Co.

<https://www.bio-techne.com/services/custom-services>

<https://www.emdgroup.com/en/news/q2-2026-06-08-2026.html>

24 Piramal process biocatalysis

Company article, 29 May 2026, describes catalogue-enzyme use and process optimisation. A bespoke-enzyme need is conditional, not established.

<https://www.piramalpharmasolutions.com/resources/blogs/biocatalysis-scalable-synthesis-pharmaceutical-manufacturing>

25 Codexis: a funded process-enzyme incumbent

Current company product description and 11 Aug 2026 results establish active commercial enzyme engineering, alongside an RNA-manufacturing strategy. Mandrake should not duplicate a catalogue.

<https://ir.codexis.com/>

<https://ir.codexis.com/news-events/press-releases/detail/452/codexis-reports-second-quarter-2026-financial-results>

26 Amgen / Generate original announcement

6 Jan 2022: five initial targets, \$50m upfront, potential \$1.9bn transaction plus royalties. Later actual payments and amendments are taken from [28].

<https://www.amgen.com/newsroom/press-releases/2022/01/amgen-and-generate-biomedicines-announce-multitarget-multimodality-research-collaboration-agreement>

27 Novartis / Generate original announcement

24 Sep 2024. \$65m upfront consideration includes \$15m equity; [28] confirms the \$50m collaboration payment and contingent structure.

<https://generatebiomedicines.com/news-releases/generatebiomedicines-announces-multi-target-collaboration-with-novartis/>

28 Generate SEC 10-Q, quarter ended 30 Jun 2026

Note 5 and MD&A collaboration agreements: actual receipts, equity, milestone caps, royalties and July 2025 amendment. Note 1 verifies IPO net proceeds. Recognised revenue is not treated as new cash.

<https://www.sec.gov/Archives/edgar/data/2100782/000119312526336437/ck0002100782-20260630.htm>

29 Source and design record

Input: Mandrake_FO_Exercise.pdf. All eleven sitemap pages, both archived research posts and linked CSS were retrieved. Fonts and palette were read from live CSS on 6 Sep 2026; dark backgrounds and print contrast adapted for this report.

<https://www.mandrake.bio/sitemap-0.xml>

https://www.mandrake.bio/_astro/global.BecZ7INs.css

https://www.mandrake.bio/_astro/platform.CvDJce6b.css

30 Current direct competition and public binder-validation inputs

Profluent announced a Lilly recombinase partnership on 28 Apr 2026. Adaptyv's public configurator lists a 60-sequence binding panel at \$10,140 with duplicate testing and 3–4 week delivery; it is a component cost, not a whole project quote. Monod markets designed binder products and partnerships. All claims remain company disclosures.

<https://www.profluent.bio/media/profluent-announces-strategic-partnership-with-eli-lilly-to-develop-ai-designed-recombinases-for-genetic-medicine>

<https://www.adaptyvbio.com/services/binding>

<https://monod.bio/novobody-2/>

31 Codexis / Merck CodeEvolver platform agreement

SEC-filed August 2015 agreement: \$5m payable shortly after signing, \$5m and \$8m transfer milestones, and up to \$15m per API in product-related payments. Field rights and use restrictions are disclosed; this is a mature-platform precedent, not Mandrake pricing.

<https://www.sec.gov/Archives/edgar/data/1200375/000120037515000019/cdxs8-kxmercklicensefinal.htm>